Experiment 5:

Calculation and Verification of Frequency Deviation and Modulation Index of FM Signals Using MATLABSimulation

Aim: To calculate the frequency deviation and modulation index of Frequency Modulated (FM) signals using MATLAB simulation.

Software Required: MATLAB

Program:

```
clc;
clear;
close all;
% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signals Using MATLAB
%% Step 1: Define parameters
Fs = 100000;            % Sampling frequency (Hz)
T = 0.02;               % Simulation duration (seconds)
t = 0:1/Fs:T-1/Fs;      % Time vector

Ac = 5;                 % Carrier amplitude (V)
Am = 2;                 % Message amplitude (V)
fc = 10000;             % Carrier frequency (Hz)
fm = 500;               % Message frequency (Hz)
kf = 500;               % Frequency sensitivity (Hz/V)
%% Step 2: Calculate frequency deviation and modulation index
delta_f = kf * Am;      % Peak frequency deviation (Hz)
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beta = delta_f / fm;     % FM modulation index
%% Step 3: Generate message and carrier signals
message = Am * cos(2*pi*fm*t);
carrier = Ac * cos(2*pi*fc*t);
%% Step 4: Generate FM signal
% FM phase = 2*pi*fc*t + beta*sin(2*pi*fm*t)
FM_signal = Ac * cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
%% Step 5: Calculate theoretical instantaneous frequency
instant_freq = fc + delta_f*cos(2*pi*fm*t);
%% Step 6: Verify frequency deviation numerically
% Estimate instantaneous frequency from unwrapped signal phase.
analytic_signal = hilbert(FM_signal);
unwrapped_phase = unwrap(angle(analytic_signal));
estimated_freq = [diff(unwrapped_phase)*Fs/(2*pi), NaN];
% Ignore initial/final edge samples for numerical verification
valid = 101:(length(t)-100);
estimated_max_freq = max(estimated_freq(valid));
estimated_min_freq = min(estimated_freq(valid));
estimated_deviation = ...
  (estimated_max_freq-estimated_min_freq)/2;

%% Step 7: Display results
fprintf('\nFM FREQUENCY DEVIATION AND MODULATION INDEX\n');
```

```
fprintf('-----------------------------------------\n');
fprintf('Carrier frequency (fc)      = %.2f Hz\n', fc);
fprintf('Message frequency (fm)      = %.2f Hz\n', fm);
fprintf('Message amplitude (Am)      = %.2f V\n', Am);
fprintf('Frequency sensitivity (kf)   = %.2f Hz/V\n', kf);
fprintf('Theoretical frequency deviation = %.2f Hz\n', delta_f);
fprintf('Theoretical modulation index    = %.2f\n', beta);
fprintf('Theoretical minimum frequency  = %.2f Hz\n', fc-delta_f);
fprintf('Theoretical maximum frequency  = %.2f Hz\n', fc+delta_f);
fprintf('\nNUMERICAL VERIFICATION\n');
fprintf('----------------------\n');
fprintf('Estimated minimum frequency = %.2f Hz\n', ...
  estimated_min_freq);
fprintf('Estimated maximum frequency = %.2f Hz\n', ...
  estimated_max_freq);
fprintf('Estimated frequency deviation = %.2f Hz\n', ...
  estimated_deviation);
fprintf('Theoretical deviation = %.2f Hz\n', delta_f);
fprintf('Simulation completed successfully.\n');
%% Step 8: Plot message, carrier, and FM signal
figure('Name','FM Signal Analysis','NumberTitle','off');
nPlot = round(0.006*Fs);
tp = t(1:nPlot);
```

```
subplot(3,1,1);
plot(tp, message(1:nPlot),'b','LineWidth',1.2);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(3,1,2);
plot(tp, carrier(1:nPlot),'k');
grid on;
title('Unmodulated Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(3,1,3);
plot(tp, FM_signal(1:nPlot),'r');
grid on;
title('Frequency Modulated (FM) Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
%% Step 9: Plot instantaneous frequency
figure('Name','Instantaneous Frequency Verification', ...
    'NumberTitle','off');
plot(t, instant_freq, 'b', 'LineWidth', 1.5);
hold on;
```

```
plot(t, estimated_freq, 'r--', 'LineWidth', 1);
yline(fc, 'k:');
hold off;
grid on;
title('Theoretical and Estimated Instantaneous Frequency');
xlabel('Time (s)');
ylabel('Frequency (Hz)');
legend('Theoretical Frequency', ...
    'Estimated Frequency', 'Carrier Frequency');

%% Step 10: Plot FM frequency spectrum
N = length(FM_signal);
Y = fft(FM_signal);
P2 = abs(Y/N);
P1 = P2(1:floor(N/2)+1);
if mod(N,2) == 0
  P1(2:end-1) = 2*P1(2:end-1);
else
  P1(2:end) = 2*P1(2:end);
end
f = Fs*(0:floor(N/2))/N;
figure('Name','FM Frequency Spectrum','NumberTitle','off');
plot(f/1000, P1, 'm', 'LineWidth', 1.2);
```

```
grid on;

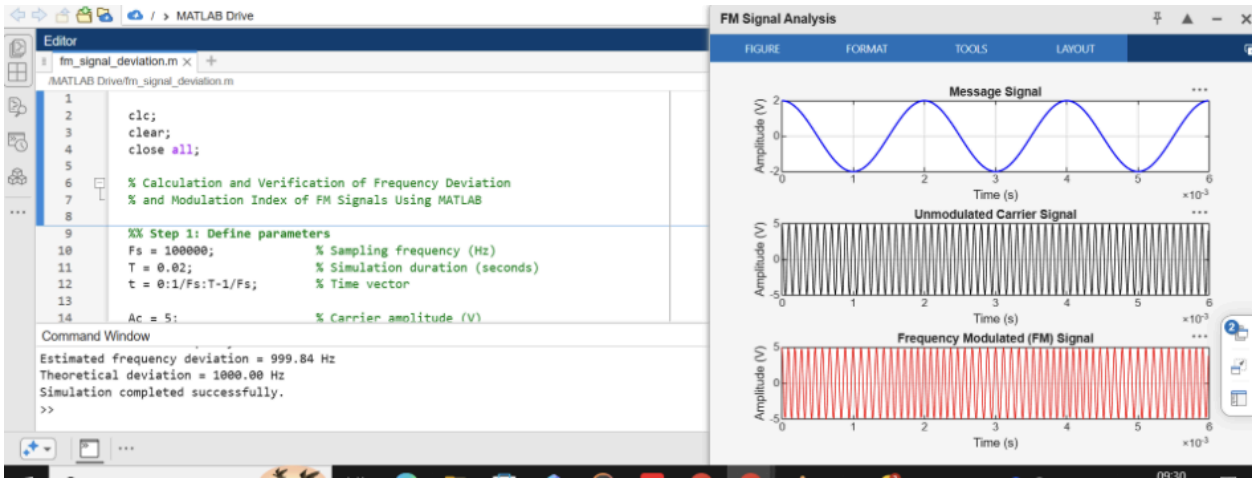
title('Frequency Spectrum of FM Signal');

xlabel('Frequency (kHz)');

ylabel('Amplitude');

xlim([5 15]);
```

Output:



Result: The FM signal was successfully generated and analyzed using MATLAB. The frequency deviation and modulation index were calculated and verified through simulation.